2.4 Sensor Trade-off Analysis

2.4.1 Distance Detection Sensor

Sensor	Advantages	Limitation
EV3 Ultrasonic	Uses sound waves to measure distance up to 255 cm.	Cannot accurately detect distances below 5 cm.
EV3 Infrared	Uses infrared light; supports beacon tracking and remote control.	Proximity detection range is approximately 50–70 cm.

- The EV3 Ultrasonic Sensor was selected instead of the EV3 Infrared Sensor because it provides direct distance measurement up to 255 cm, making it more suitable for detecting obstacles and maintaining a safe distance during navigation.
- Although the Infrared Sensor supports beacon tracking and remote control, its proximity detection range of approximately 50–70 cm is less suitable for our requirements.
- The Ultrasonic Sensor was chosen to provide more suitable distance feedback for obstacle avoidance.

2.4.2 Colour Detection Sensor

Sensor	Advantages	Limitation
NXT Colour Sensor	Detects 6 colours; approximately 333 Hz sampling rate.	Fewer detectable colours and lower sampling rate.
EV3 Colour Sensor	Detects 7 colours; includes Reflected Light mode; up to 1000 Hz sampling rate.	Sensitive to lighting conditions and sensor distance.

- The EV3 Colour Sensor was selected instead of the NXT Colour Sensor because it supports more colour detection options (7 colours compared to 6) and has a higher sampling rate of up to 1000 Hz compared to approximately 333 Hz.
- This provides faster data acquisition and greater sensing capabilities for detecting field markings.
- The EV3 Colour Sensor was chosen for more responsive and reliable colour detection.

2.4.3 Visual Recognition Camera

Sensor	Advantages	Limitation
Pixy Camera	Built-in colour recognition simplifies visual processing.	Sensitive to lighting and similar colours.
OpenMV Camera	Supports flexible and advanced image processing.	Complex integration; transmission errors occurred during testing.

- The Pixy Camera was selected instead of the OpenMV Camera because it provides built-in colour recognition, making it more practical for detecting red and green pillars.
- In comparison, the Pixy Camera offered a simpler integration process and was suitable for our pillar recognition requirements.
- Therefore, the Pixy Camera was selected to support more reliable visual detection during navigation.

2.4.4 Heading Detection – With Gyro vs Without Gyro

Sensor	Advantages	Limitation
With Gyro	Simple system; fewer components and less programming complexity.	Relies on predetermined motor rotations; turning accuracy may be affected by wheel slippage and mechanical variations.
Without Gyro	Provides real-time heading feedback; allows turning corrections and improves navigation consistency.	May experience gyro drift, calibration errors and inaccurate readings during operation.

- The EV3 Gyro Sensor was selected instead of relying solely on motor rotation to control turning angles. Without a Gyro Sensor, the robot depends on predetermined motor rotations, which may result in inaccurate turning due to wheel slippage.
- With the Gyro Sensor, the robot can monitor its actual heading and adjust its steering to achieve the desired turning angle.
- Therefore, the Gyro Sensor was selected to improve turning accuracy and navigation consistency.